

Walkalong Glider

An Investigation of Aerodynamics and
Stability of Low Reynold Number Flight

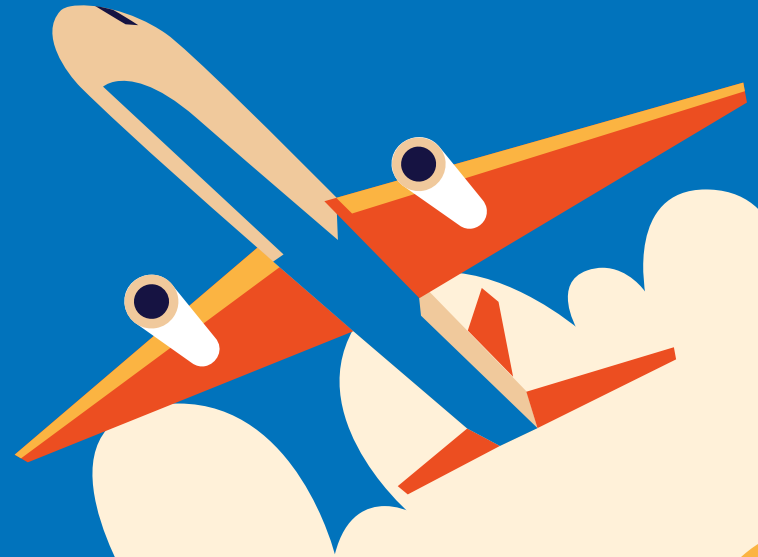


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Definition

A Walkalong Glider is defined as a lightweight, ***unpowered airfoil*** which can ***sustain flight*** within a ***user-generated flow field*** maintained at a ***walking pace***.

Unforced

$$\begin{aligned}\dot{u} &= wq + \frac{1}{m} [X_0 - mg \sin \theta], \\ \dot{w} &= -uq + \frac{1}{m + M_{33}} [Z_0 + mg \cos \theta], \\ \dot{q} &= \frac{1}{I_{\text{eff}}} M_0, \\ \dot{x}_g &= u \cos \theta - w \sin \theta, \\ \dot{z}_g &= u \sin \theta + w \cos \theta, \\ \dot{\theta} &= q.\end{aligned}$$

$$X_0 = -D_0 \cos \alpha_0 + L_0 \sin \alpha_0,$$

$$Z_0 = -L_0 \cos \alpha_0 - D_0 \sin \alpha_0,$$

$$L_0 = \frac{1}{2} \rho_f S V_0^2 C_L(\alpha_0),$$

$$D_0 = \frac{1}{2} \rho_f S V_0^2 C_D(\alpha_0).$$

$$V_0 = \sqrt{u^2 + w^2}, \quad \alpha_0 = \theta + \tan^{-1} \left(\frac{w}{u} \right).$$

$$\begin{aligned}u_{\text{rel}} &= u - u_{\text{ind}}^B, & w_{\text{rel}} &= w - w_{\text{ind}}^B, \\ V_{\text{rel}} &= \sqrt{u_{\text{rel}}^2 + w_{\text{rel}}^2}, & \alpha_{\text{rel}} &= \theta + \tan^{-1} \left(\frac{w_{\text{rel}}}{u_{\text{rel}}} \right), \\ L_{\text{rel}} &= \frac{1}{2} \rho_f S V_{\text{rel}}^2 C_L(\alpha_{\text{rel}}), & D_{\text{rel}} &= \frac{1}{2} \rho_f S V_{\text{rel}}^2 C_D(\alpha_{\text{rel}}).\end{aligned}$$

Forced

$$\begin{aligned}\dot{u} &= wq + \frac{1}{m} [X_0 + \Delta X_w - mg \sin \theta], \\ \dot{w} &= -uq + \frac{1}{m + M_{33}} [Z_0 + \Delta Z_w + mg \cos \theta], \\ \dot{q} &= \frac{1}{I_{\text{eff}}} [M_0 + \Delta M_w], \\ \dot{x}_g &= u \cos \theta - w \sin \theta, \\ \dot{z}_g &= u \sin \theta + w \cos \theta, \\ \dot{\theta} &= q, \\ \dot{x}_w &= V_w(t).\end{aligned}$$

$$\Delta X_w = X_{\text{rel}} - X_0,$$

$$\Delta Z_w = Z_{\text{rel}} - Z_0,$$

$$\Delta M_w = M_{\text{rel}} - M_0.$$

$$X_{\text{rel}} = -D_{\text{rel}} \cos \alpha_{\text{rel}} + L_{\text{rel}} \sin \alpha_{\text{rel}},$$

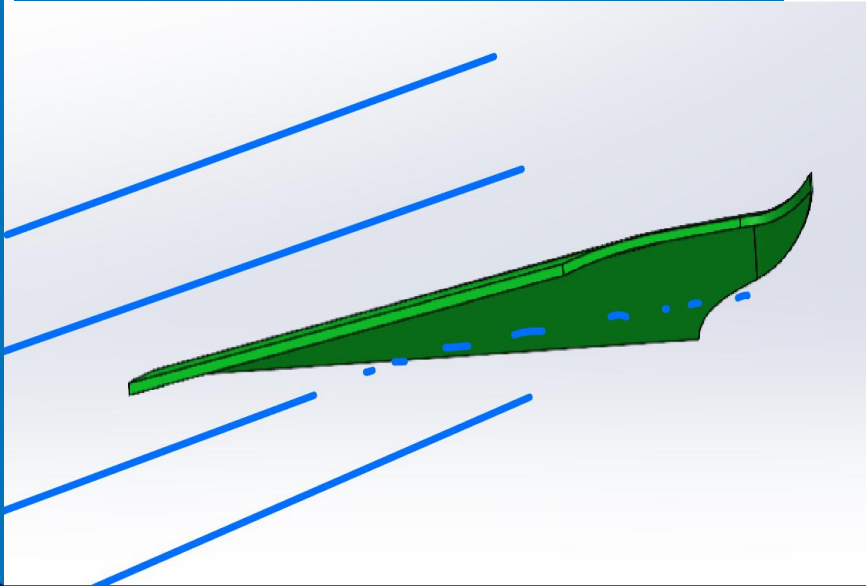
$$Z_{\text{rel}} = -L_{\text{rel}} \cos \alpha_{\text{rel}} - D_{\text{rel}} \sin \alpha_{\text{rel}}.$$

Forced case = unforced dynamics + walker-induced aerodynamic increments.

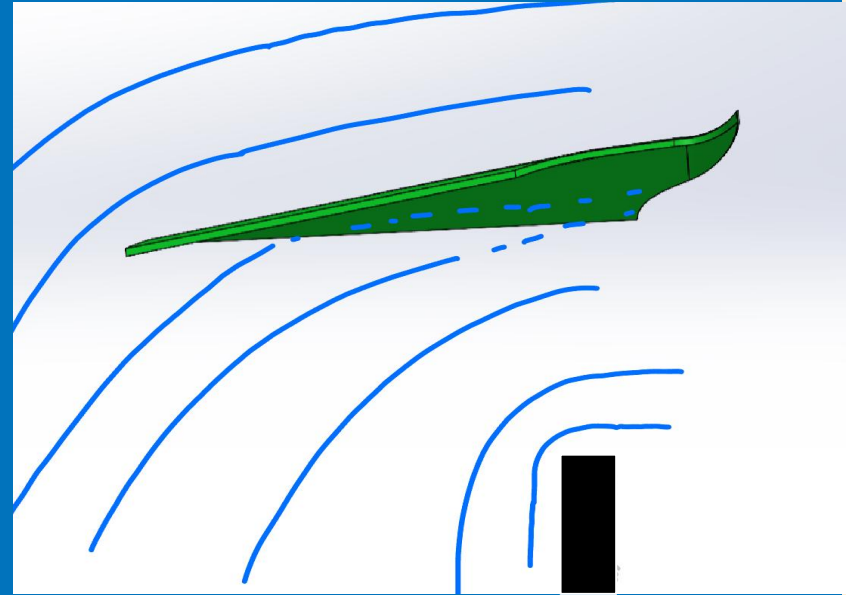
Theory

Explanation with
diagram

Unforced

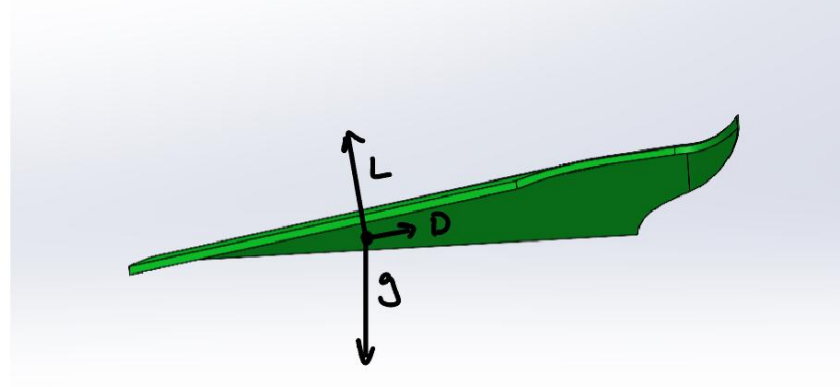


Forced



Theory

Lift and Drag Coefficients, briefly



$$L \propto \frac{1}{2} \rho v^2 S$$

$$D \propto \frac{1}{2} \rho v^2 S$$

$$L = mg \cos \gamma$$

$$D = mg \sin \gamma$$



System Categorization

What is Unique about a
Walkalong Glider?

Low Re Flight



Birds

$Re \sim 10^5$



**Commercial
Planes**

$Re \sim 10^6 - 10^7$



Our Glider

$Re \sim 10^3 - 10^4$

Walkalong Gliders sit right on the transition from laminar to turbulent flow, making flow unsteady and incredibly sensitive to changes in geometry, fluctuations, etc.

How is steady lift achieved in this regime?

Design Considerations

Weight

- EPS foam

Roll Stability

- Dihedral
- Low CoM

Pitch Stability

- Forward CoM
- Chord Length
- Trailing Edge
Camber

Separation Reduction

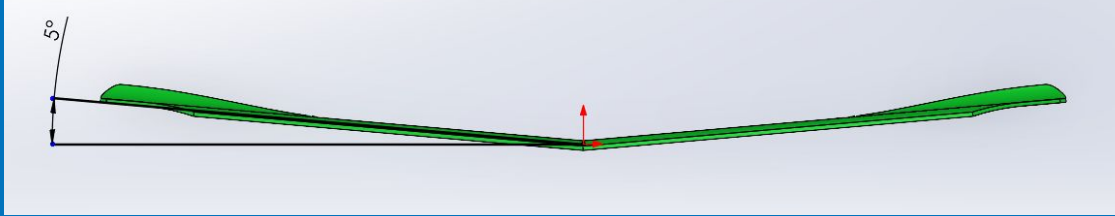
- Local AoA
adjustment

Weight

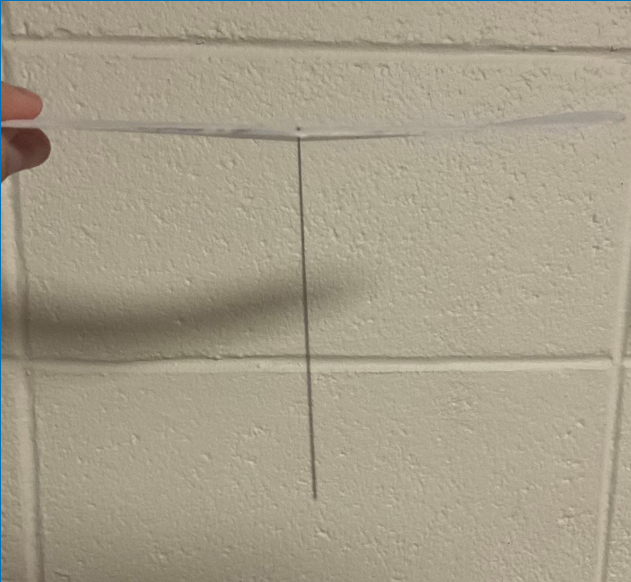
- Lowest Weight possible to keep it slow/walkable
- Low inertia so a person can push it
- recurring theme



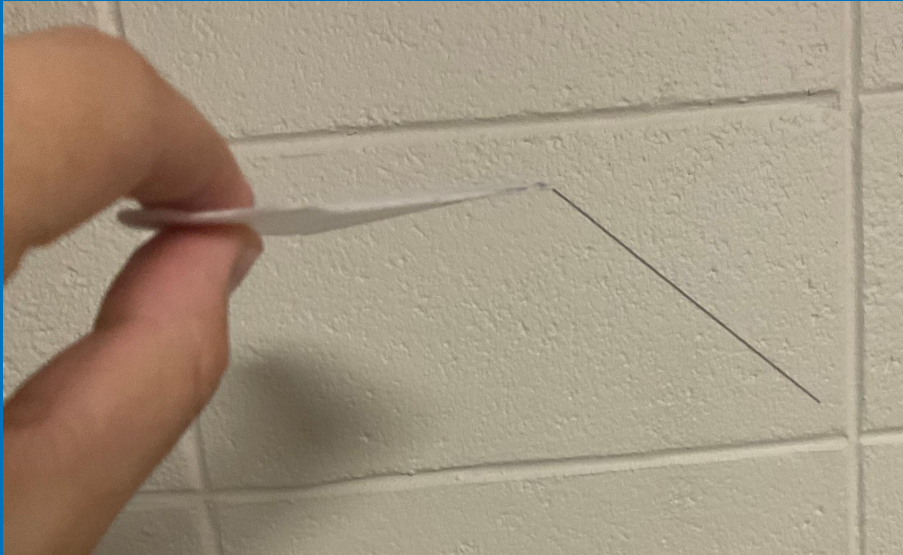
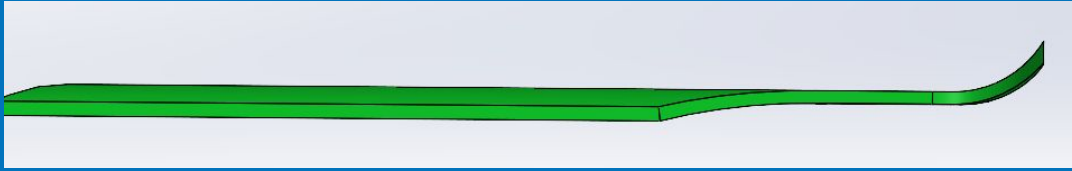
Roll Stability



- Dihedral
 - angled wings allows for roll stability by adjusting lift on wings during roll
- Low CoM
 - CoM acts like pendulum: lowest frequency when CoM is lower

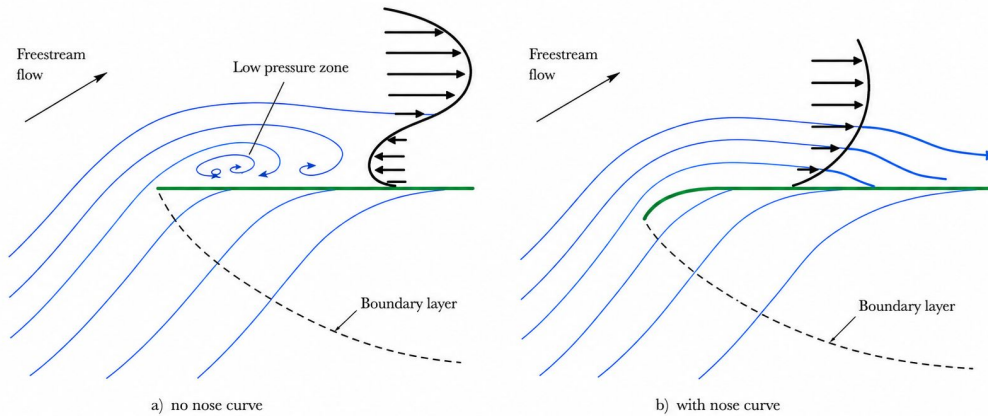


Pitch Stability



- Trailing Edge Camber
 - Produces torque about CoM to keep positive AoA
- Forward CoM
 - Increase restoring torque change in AoA

Separation Reduction

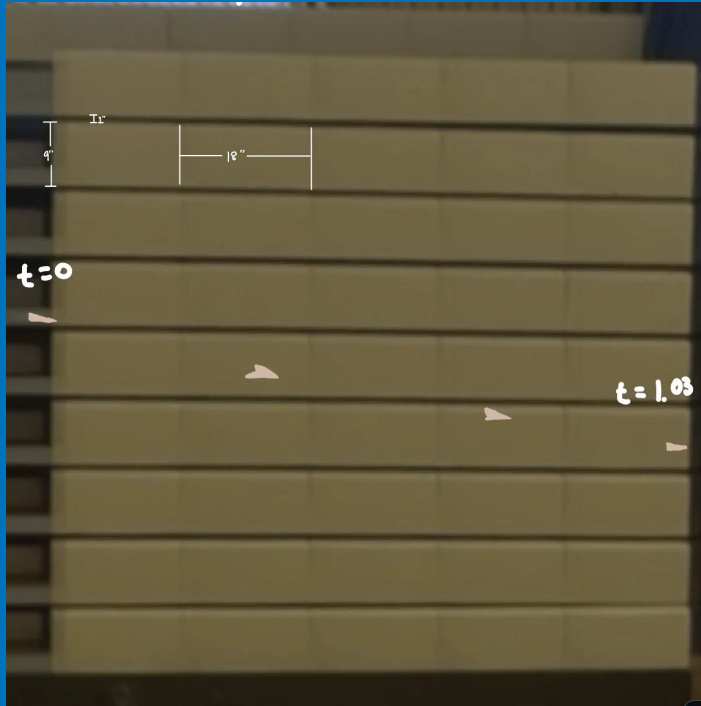


- Keep flow attached to reduce separation, flutter
 - local AoA aligns with flow

Experimental Results – We Did It!



Experimental Results



- Mass: $m = 0.94 \text{ g}$
- glide ratio: $\tan(\gamma) = 18''/90'' = 0.2$
- Glide speed: $V = 2.2 \text{ m/s}$
- Projected area: $S = 0.011 \text{ m}^2$
- Mean Chord: $c = 0.0635 \text{ m}$

Dimensionless Coefficients

- $C_L = 0.265$
- $C_D = 0.0531$
- $Re \sim 9.4 \times 10^3$



Experimental Results

Compared to Heavier model

- Mass: $m = 1.68 \text{ g}$
- glide ratio: $\tan(\gamma) = 18''/90'' = 0.25$
- Glide speed: $V = 2.8 \text{ m/s}$
- Projected area: $S = 0.011 \text{ m}^2$
- Mean Chord: $c = 0.0635 \text{ m}$

- $C_L = 0.558$
- $C_D = 0.140$
- $Re = 1.2 \times 10^4$

The heavier version had steeper angle of attack and glided faster than the lighter version. This made it impossible to fly at a walking pace and required more upwash to fly which is why a high surface object was needed to be held to fly it.





References

- ScienceToyMaker: <http://sciencetoymaker.org>
 - Website and youtube channel
 - Alblas, Dieuwertje & van den Bosch, Mark & Feigl, Zoro & Klomp, Lucas & Rottschäfer, Vivi & Schwenninger, Felix & Spek, Len & Weedage, Lotte & Zeijlemaker, Sjanne. (2023). A Joint Data- and Model-Driven Approach Simulating the Behaviour of a Walkalong Glider. 10.33774/miir-2023-p957p.
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Thank you!

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